



The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure (years)

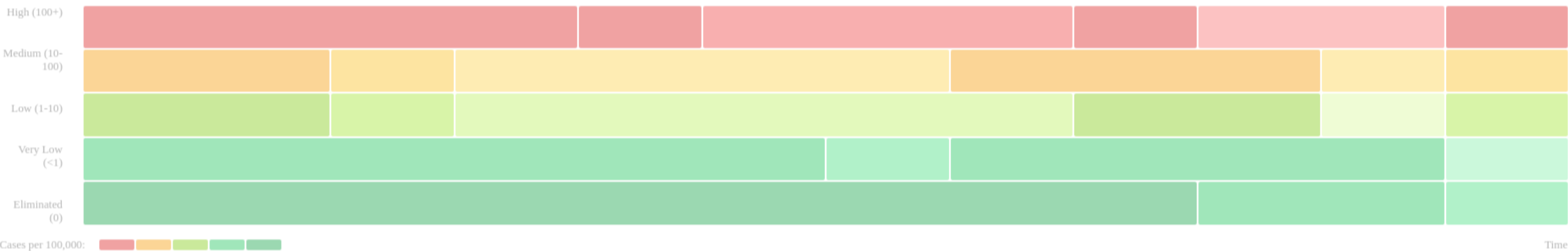


100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

DISEASE	PRE-VACCINE (PER MILLION/YR)	POST-VACCINE (PER MILLION/YR)	CASE REDUCTION	DEATH REDUCTION
Diphtheria	158	0	100%	100%
Smallpox	250	0	100%	100%
Acute Polio	141	0	100%	100%
Measles	3,044	0.2	99.99%	100%
Pertussis	1,534	52	96.6%	99.7%
Hepatitis A	465	51	89%	88.7%
Acute Hepatitis B	273	44	83.9%	83.6%
Pneumococcal	233	139	40.5%	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles Incidence by State (per 100,000)



Smallpox: The Only Human Disease Eradicated Through Vaccination

1959

WHO Passes Resolution

World Health Organization passes resolution to eradicate smallpox globally

1967

Intensified Campaign Launched

WHO launches intensified global eradication program with enhanced surveillance and containment

1977

Last Natural Case

Ali Maow Maalin, a hospital cook in Somalia, becomes the last person with naturally acquired smallpox

1980

Certified Eradicated

WHO officially certifies smallpox eradicated — the only human disease eliminated through vaccination

WHY ERADICATION WORKED

Humans Only

Virus infected only humans — no animal reservoir

DISTINGUISHING FEATURE

Visible Symptoms

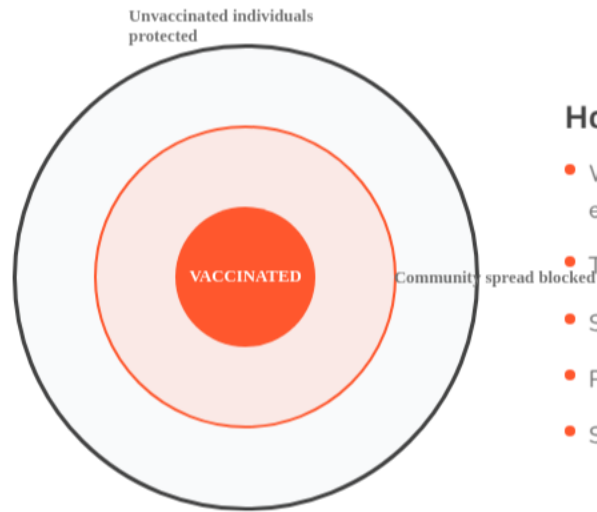
Rash was easily recognizable, enabling rapid containment

VACCINE PROPERTIES

Effective & Affordable

Single dose provided lasting immunity; needle-free delivery possible

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

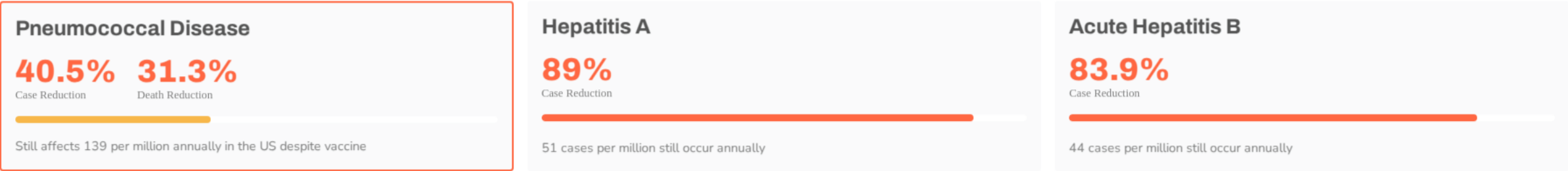


How It Works

- When a high percentage of a population is vaccinated, pathogens cannot spread effectively
- This creates indirect protection for those who cannot be vaccinated
- Shielding newborns before they can be vaccinated
- Protecting immunocompromised patients
- Safeguarding cancer patients undergoing chemotherapy



Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Key Barriers to Complete Coverage

- Access Gaps
- Supply Shortages
- Vaccine Hesitancy
- Limited Newer Vaccines
- Geographic Disparities

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1

100% Elimination Achieved
Diphtheria, smallpox, and both forms of polio eliminated in the US through vaccination
- 2

Development Timelines Collapsed
From 100+ years (typhoid) to under 1 year (COVID-19) — driven by mRNA and genomic tools
- 3

Herd Immunity Protects Communities
Population-level vaccination shields newborns, immunocompromised, and chemotherapy patients
- 4

Gaps Remain for Some Diseases
Pneumococcal (40.5%), hepatitis A (89%), and hepatitis B (83.9%) — need improved coverage
- 5

Smallpox Proved Eradication Is Possible
The only human disease eradicated through vaccination — certified by WHO in 1980

